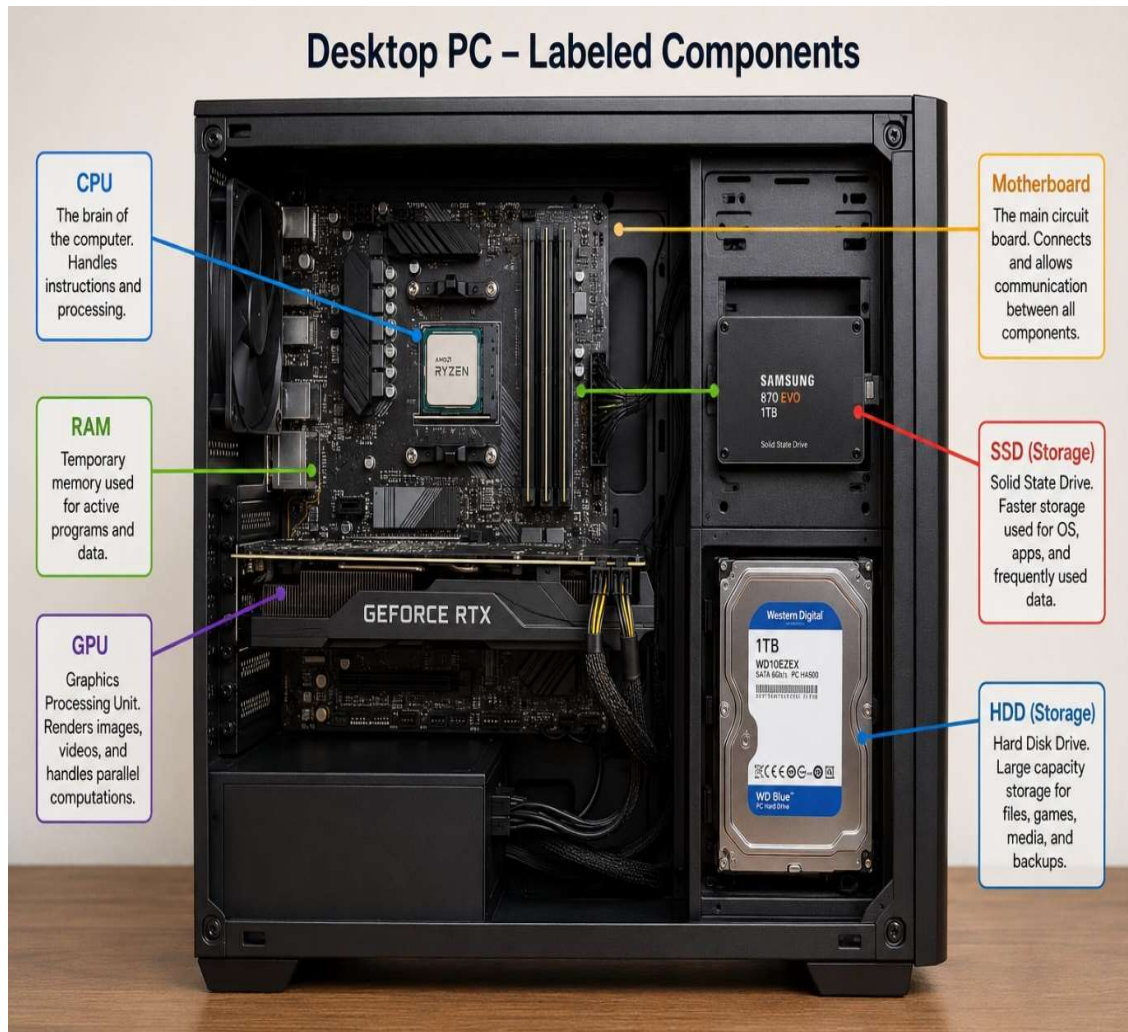


1. Draw a labelled diagram of a desktop PC, clearly marking the motherboard, CPU, RAM, GPU, and both HDD and SSD storage components.



2. Create a comparison table showing the main differences between HDD and SSD storage in terms of speed, durability, and typical use cases.

Feature / Metric	Hard Disk Drive (HDD)	Solid-State Drive (SSD)
Random Access Time	5–10 ms	0.1 ms
Read Latency Time	High	Very low
Data Transfer Rate	50 MB/s to 100 MB/s	100 MB/s to 500 MB/s
Moving Parts	Yes (spinning platters, read/write heads)	None (uses flash memory chips)
Reliability	Lower	High
Form Factor	large and heavy weight	Small and light weight
Common Capacities	High capacity (commonly up to 4TB and higher)	Typically, up to 2TB (higher capacities exist but are expensive)
Power Consumption	6–12 Watts	2 Watts

3. Write a short explanation (2-3 sentences each) describing the role of the CPU, RAM, and GPU in running a game like PUBG or Free Fire on a PC.

1. The CPU (The Brain / The Manager)

The CPU is the boss that runs the game's rules and logic. In PUBG, it calculates things you can't see right away, like where a bullet hits when you shoot, how the enemy bots move, and keeping your game in sync with other players online. Without a good CPU, your game will lag and stutter because the "boss" can't think fast enough.

2. The RAM (The Desktop / Short-Term Memory)

Think of RAM as a large desk where you keep things you need to grab instantly. When you open Free Fire, the computer copies the game's map, gun models, and sounds from your slow storage drive onto this fast desk. This allows your computer to load things instantly as you run around the map, instead of freezing every time you enter a new area.

3. The GPU / Graphics Card (The Artist)

The GPU is the artist that draws everything you actually see on your screen. The CPU gives it the raw data, and the GPU instantly paints the 3D grass, the shadows, the characters, and the explosions. A powerful GPU means the game will look beautiful, crisp, and smooth instead of looking blurry or choppy.

4. Imagine you want to upgrade your PC to run the latest version of FIFA smoothly. List which hardware components you would consider upgrading and explain why for each.

Graphics Card (GPU) — The Artist

- **What it does:** It paints everything you see on screen, like the detailed player faces, the green grass, and stadium crowds.
- **Why upgrade:** It stops the game from looking like a choppy, laggy slideshow and makes it look buttery smooth.

Processor (CPU) — The Brain

- **What it does:** It handles the rules of the game, like how the ball bounces and what the AI players do.
- **Why upgrade:** It stops the computer from freezing up when the penalty box gets crowded with too many players.

Memory (RAM) — The Workspace

- **What it does:** It acts like a giant, super-fast desk that keeps the current match files open and ready to grab.
- **Why upgrade:** Having 16GB of RAM stops the game from suddenly hitching right when you are about to shoot.

Storage (SSD) — The Speed Pass

- **What it does:** It opens up the heavy game files instantly instead of slowly searching through them like an old hard drive.
- **Why upgrade:** It completely skips painful, long loading screens so you can get straight onto the pitch to play.